

ASHVAJEET ATHAWALE

B.Tech 4th Year | Malviya National Institute of Technology, Jaipur

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EDUCATION

B.Tech in Electronics & Communication Engineering

Nov 2022 – May 2026

Malviya National Institute of Technology, Jaipur

- **Class XII (CBSE, Mar 2021):** Scored 92%.
- **Class X (CBSE, Mar 2019):** Scored 86.7%.

PUBLICATIONS & PATENTS

- **System and Method for Real-Time Neural Data Analysis** Aug 2025
 - **Patent Filed:** Co-authored and filed an invention patent named as "Multi-Band electromagnetic burst analyzer and polyatomic time-crystal processor". (*Status: Pending*)
 - **Research Paper:** Co-authored paper "Beyond Conventional EEG: High-Frequency Circuits Design and Brain Data Visualization for DDG device," in preparation for submission to Journal of Applied Physics.

WORK EXPERIENCE

- **Research Intern** May 2025 - Jul 2025
IIT Mandi
 - Delivered the world's first experiment-ready prototype for high-frequency (up to 5.31 GHz) brain scanning device, featuring 4 innovative real-time animating neural visualizations for doctors & researchers.
 - Architected the C++ backend for 10Gbps/168-channel FFT processing with hardware integration from scratch and neural data mapping with heatmaps and polyatomic time crystal using HTML canvas
- **Full Stack Developer Intern** May 2024 - Feb 2024
Per Clean Laundry Services
 - Engineered different APIs, including a React.js report generation module, optimized query handling, OTP verification and Whatsapp notification, and backend for 30+ APIs, with advanced logger debugging system
 - Developed a custom unit testing framework from scratch using Node.js, without using external libraries, and developed the LLD document solution for all orders fetching improving latency of 40%.

PROJECTS

- **Automated Neural State Classification from DDG Visualizations** Aug 2024
 - Developed a Computer Vision model using OpenCV and TensorFlow to classify neural states from 4 distinct real-time visualizations, processing data stream from 168 channels.
 - Automated the analysis process, reducing pattern identification time by over 90% compared to manual analysis and enabling more efficient quantification of neural activity.
- **Campus Prep | AI-Powered Learning Analysis** Jul 2024
 - Developed a hybrid architecture using BERT and LSTM for contextual and sequential keystroke analysis, with a multi-head attention block reducing training time from 6 hours to 30 min per batch.
 - Integrated a data pipeline in Next.js platform capturing over 500 real-time metrics such as syntax errors from an integrated code editor with an actionable, AI-generated roadmap and strengths and weakness.
- **Staged Memory Transformer Model (SMT)** Jun 2024
 - Devised a hierarchical attention model with staged memory tokens to cut computational complexity from $O(n^2)$ to $O(n)$, enabling a 10,000-token sequence length on a single 4GB GPU.
 - Validated the architecture to achieve a 40% reduction in GPU training time and a 2.5x increase in long-context retention compared to standard transformer models.

SKILLS

- **AI & Computer Vision:** Machine Learning, Deep Learning, Computer Vision, Image Processing
- **Libraries & Frameworks:** OpenCV, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib, Jest
- **Programming:** Python, C++, JavaScript, TypeScript, SQL
- **Developer Tools:** Git, GitHub, Docker, Postman, Google colab, MongoDB compass
- **Services:** AWS, Google Cloud, Twilio, Logger
- **Relevant Coursework:** Neural Network, Deep Learning, Computer Vision, Image Processing by Coursera , React.js with Redux Toolkit by Udemy

POSITION OF RESPONSIBILITY

- **General Secretary, Classical Music & Dance Club:** Led club team for 'Nratyanjali' and 'Voice Toice' events during BLITZSCHLAG '24 (Cultural Fest).
- **Joint Secretary, Aeromodelling Club:** Head of 'Aeroquest', the flagship event of SPHINX '24 (Technical Fest).